

Machine Learning



Module 1

Introduction to Machine Learning

Module 1:

Introduction to Machine Learning

- Types of human learning.
- What is machine learning?
- Types of machine learning
 - ❖ Supervised
 - ❖ Unsupervised
 - ❖ Semi-supervised
 - ❖ Reinforcement learning
- Machine learning activities
- Applications of machine learning.



Human Learning

- In cognitive science, learning is typically referred to as the process of gaining information through observation.
- To do a task in a proper way, we need to have prior information on one or more things related to the task.
- Also, as we keep learning more or in other words acquiring more information, the efficiency in doing the tasks keep improving.
- Thus, with more learning, tasks can be performed more efficiently.

Types of human learning

- Human learning happens in one of the three ways –
 - (1) Either somebody who is an expert in the subject directly teaches us.
 - (2) We build our own notion indirectly based on what we have learnt from the expert in the past, or
 - (3) We do it ourselves, may be after multiple attempts, some being unsuccessful.

(1) Learning under expert guidance

- In all phases of life of a human being, there is an element of guided learning.
- This learning is imparted by someone, purely because of the fact that he/she has already gathered the knowledge by virtue of his/her experience in that field.
- So guided learning is the process of gaining information from a person having sufficient knowledge due to the past experience.

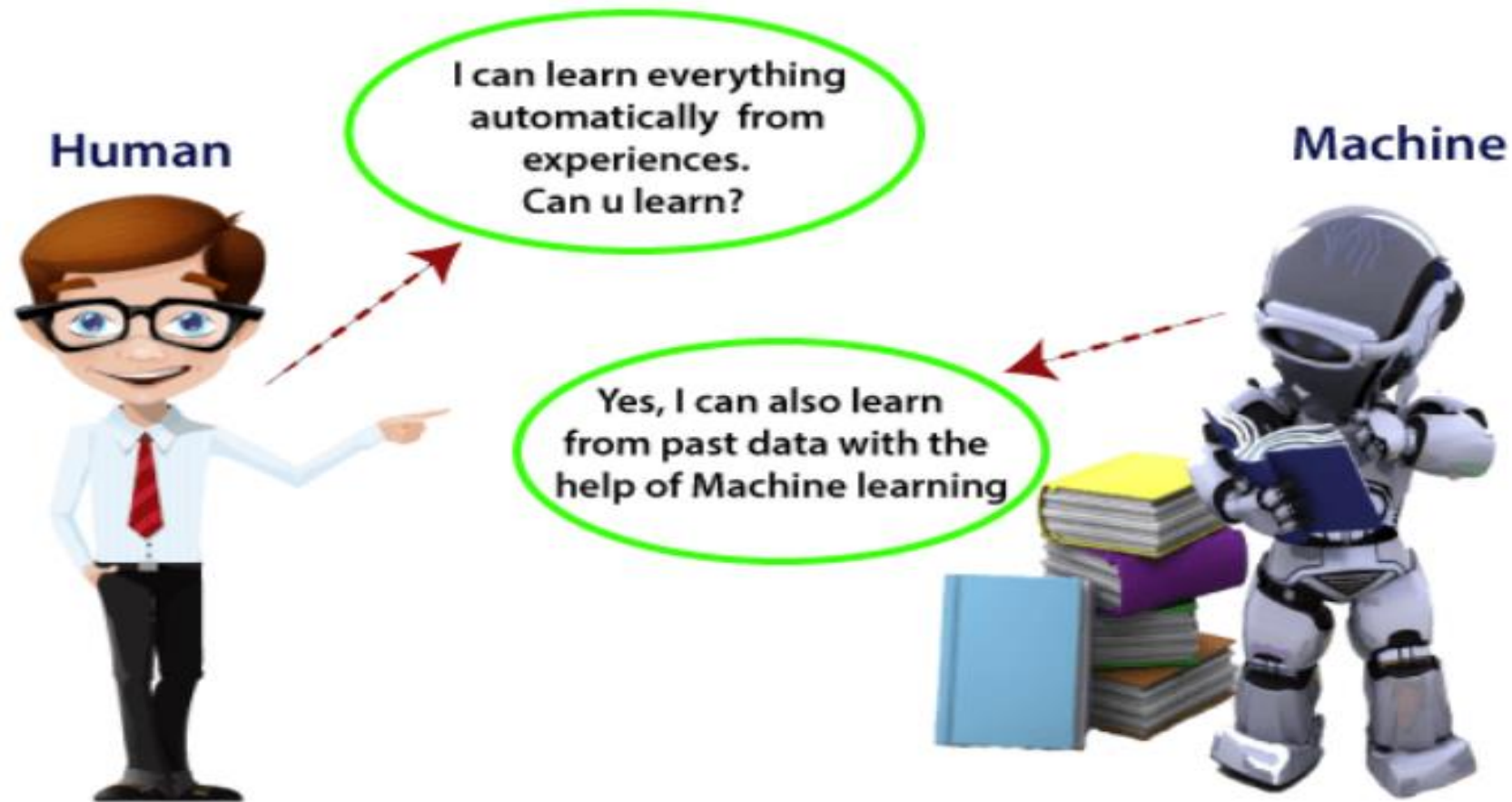
(2) Learning guided by knowledge gained from experts

- There is no direct learning.
- It is some past information shared on some different context, which is used as a learning to make decisions.

(3) Learning by self

- Not all things are taught by others. A lot of things need to be learnt only from mistakes made in the past.
- We tend to form a check list on things that we should do, and things that we should not do, based on our experiences.

What is machine learning?



Machine learning

- Subset of **artificial intelligence** (AI).
- It is focused on teaching computers to learn from data and to improve with experience – instead of being explicitly programmed to do so.
- According to **Tom M. Mitchell**, Professor of Machine Learning Department, School of Computer Science, Carnegie Mellon University, Machine learning is:
- **‘A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P, if its performance at tasks in T, as measured by P, improves with experience E.’**

Example:

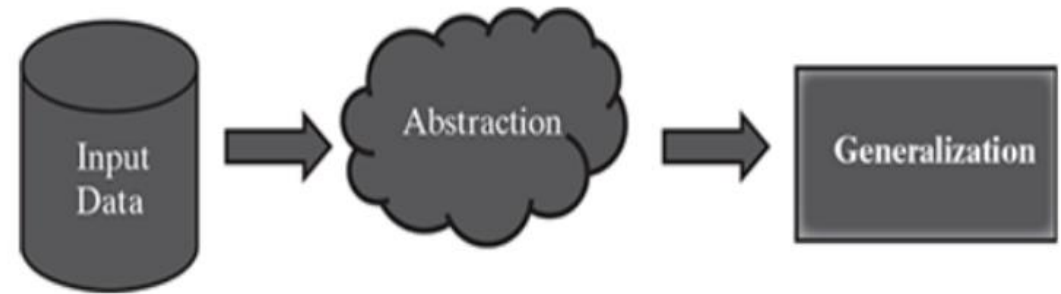
- In context of image classification,
- E represents the past data with images having labels or assigned classes (for example whether the image is of a class cat or a class dog or a class elephant etc.)
- T is the task of assigning class to new, unlabeled images and
- P is the performance measure indicated by the percentage of images correctly classified.
- The first step in any project is defining your problem. Even if the most powerful algorithm is used, the results will be meaningless if the wrong problem is solved.

How do machines learn?

The basic machine learning process can be divided into three parts.

- 1. Data Input:** Past data or information is utilized as a basis for future decision-making
- 2. Abstraction:** The input data is represented in a broader way through the underlying algorithm
- 3. Generalization:** The abstracted representation is generalized to form a framework for making decisions

Fig: Schematic representation of the machine learning process.



Abstraction

- Abstraction helps in deriving a conceptual map based on the input data.
- This **map, or a model** as it is known in the machine learning paradigm, is summarized knowledge representation of the raw data.
- The model may be in any one of the following forms
 - Computational blocks like if/else rules
 - Mathematical equations
 - Specific data structures like trees or graphs
 - Logical groupings of similar observations

- The choice of the model used to solve a specific learning problem is a human task.
- The decision related to the choice of model is taken based on multiple aspects, some of which are listed below:
 - ❖ **The type of problem to be solved:** Whether the problem is related to forecast or prediction, analysis of trend, understanding the different segments or groups of objects, etc.
 - ❖ **Nature of the input data:** How exhaustive the input data is, whether the data has no values for many fields, the data types, etc.
 - ❖ **Domain of the problem:** If the problem is in a business critical domain with a high rate of data input and need for immediate inference, e.g. fraud detection problem in banking domain.

- Once the model is chosen, the **next task is to fit the model** based on the input data. Let's understand this with an example.
- In a case where the model is represented by a mathematical equation, say ' $y = c_1 + c_2 x$ ' (the model is known as simple linear regression), based on the input data, we have to find out the values of c_1 and c_2 . Otherwise, the equation (or the model) is of no use.
- So fitting the model, in this case, means finding the values of the unknown coefficients or constants of the equation or the model.
- This process of fitting the model based on the input data is known as **training**. Also, the input data based on which the model is being finalized is known as training data.

Generalization

- Abstraction process, or more popularly training the model, is just one part of machine learning.
- The other key part is to tune up the abstracted knowledge to a form which can be used to take future decisions. This is achieved as a part of **generalization**. This part is quite difficult to achieve.
- This is because the model is trained based on a finite set of data, which may possess a limited set of characteristics.

- But when we want to apply the model to take decision on a set of unknown data, usually termed as test data, we may encounter two problems:
 1. The trained model is aligned with the training data too much, hence may not portray the actual trend.
 2. The test data possess certain characteristics apparently unknown to the training data.
- Hence, a precise approach of decision-making will not work. An approximate or heuristic approach, much like gut-feeling-based decision-making in human beings, has to be adopted.

Well-posed learning problem

- For defining a new problem, which can be solved using machine learning, a simple framework, highlighted below, can be used.
- This framework also helps in deciding whether the problem is a right candidate to be solved using machine learning. The framework involves answering three questions:
 - 1. What is the problem?**
 - 2. Why does the problem need to be solved?**
 - 3. How to solve the problem?**

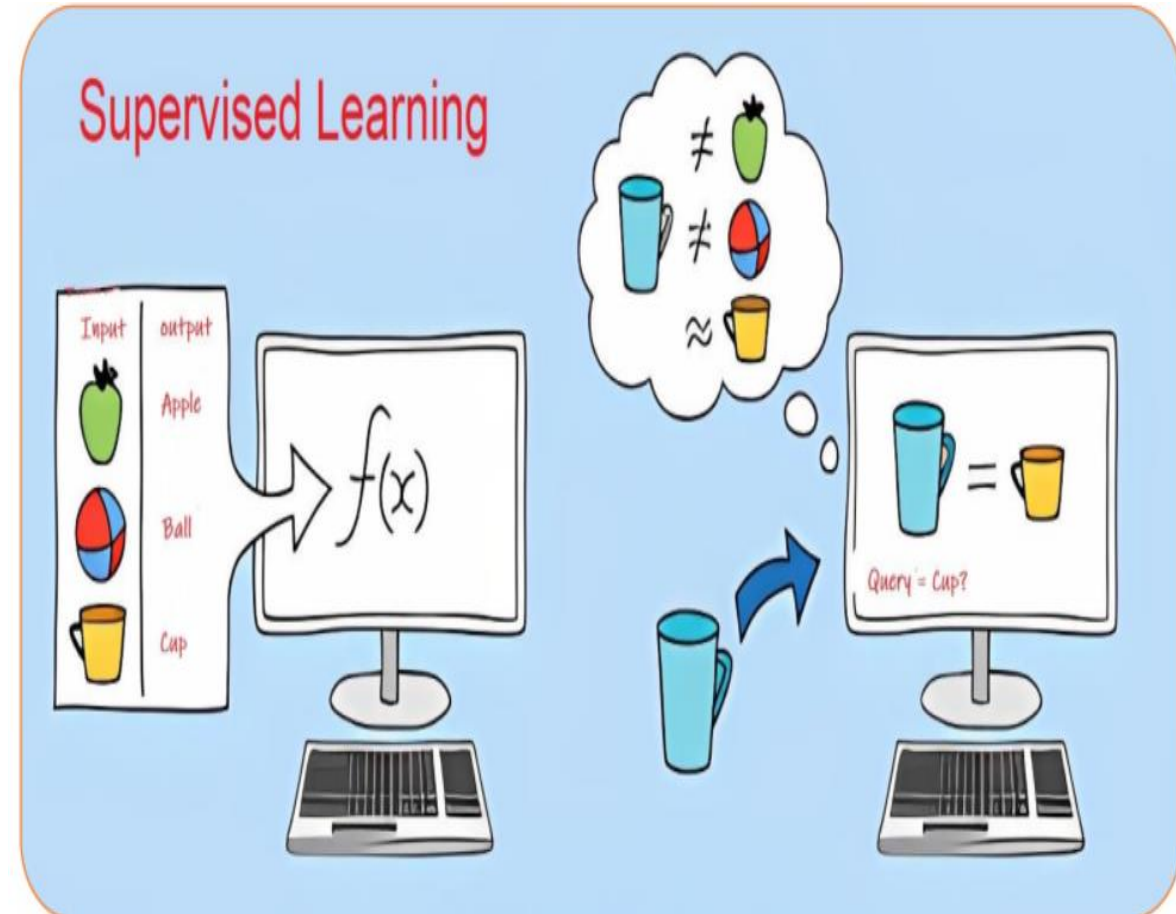
- **Step 1: What is the problem?** Describe the problem informally and formally and list assumptions and similar problems.
- **Step 2: Why does the problem need to be solved?** List the motivation for solving the problem, the benefits that the solution will provide and how the solution will be used.
- **Step 3: How would I solve the problem?** Describe how the problem would be solved manually to flush domain knowledge.

Types of learning

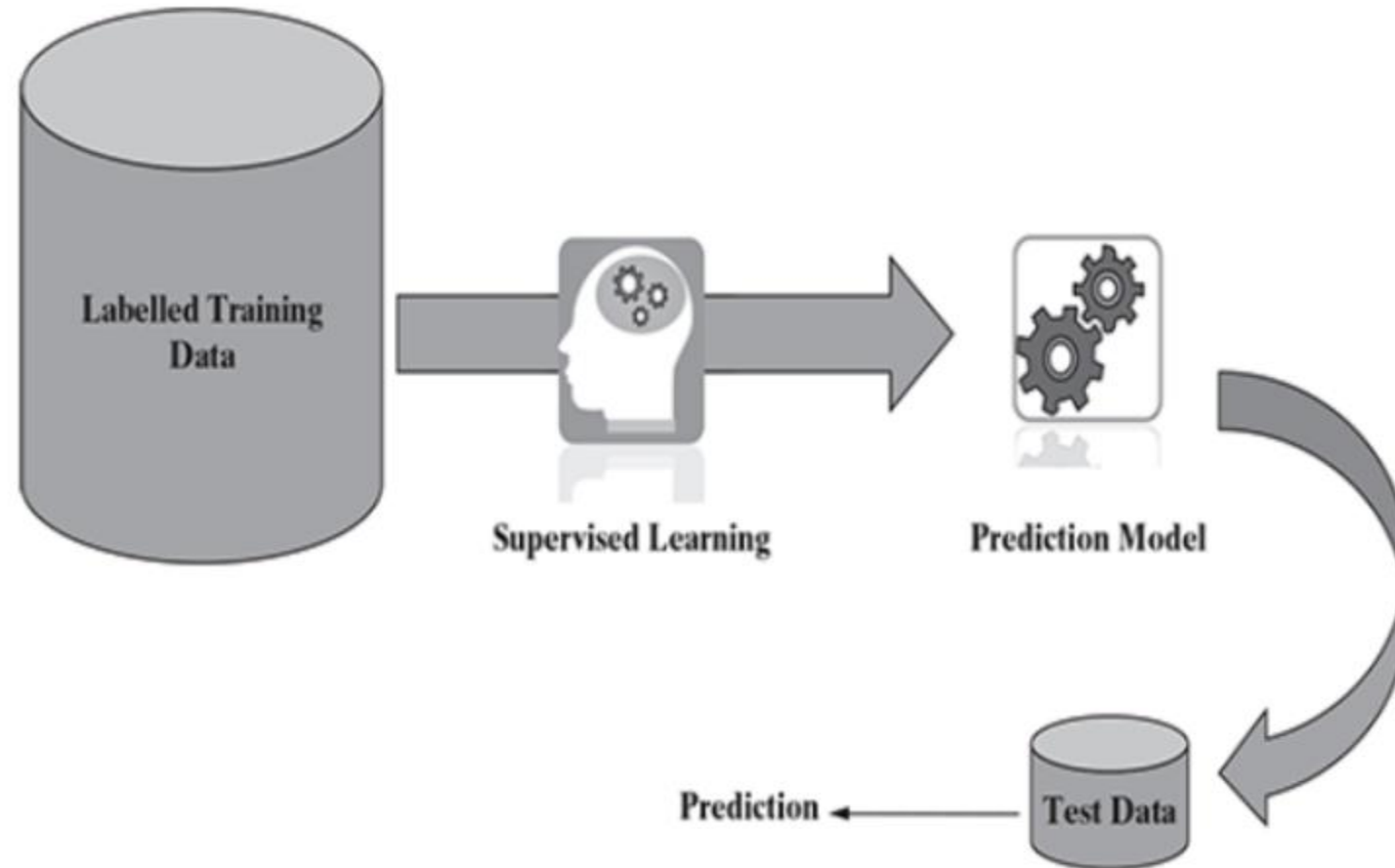
1. **Supervised learning** – Also called predictive learning. A machine predicts the class of unknown objects based on prior class related information of similar objects.
2. **Unsupervised learning** – Also called descriptive learning. A machine finds patterns in unknown objects by grouping similar objects together.
3. **Semi-Supervised learning** -Represents the intermediate ground between Supervised and Unsupervised learning algorithms. It uses the combination of labeled and unlabeled datasets during the training period.
4. **Reinforcement learning** – A machine learns to act on its own to achieve the given goals.

Supervised Learning

Supervised learning is the machine learning task of learning a function that maps an input to an output based on example input-output pairs.



Supervised learning Process



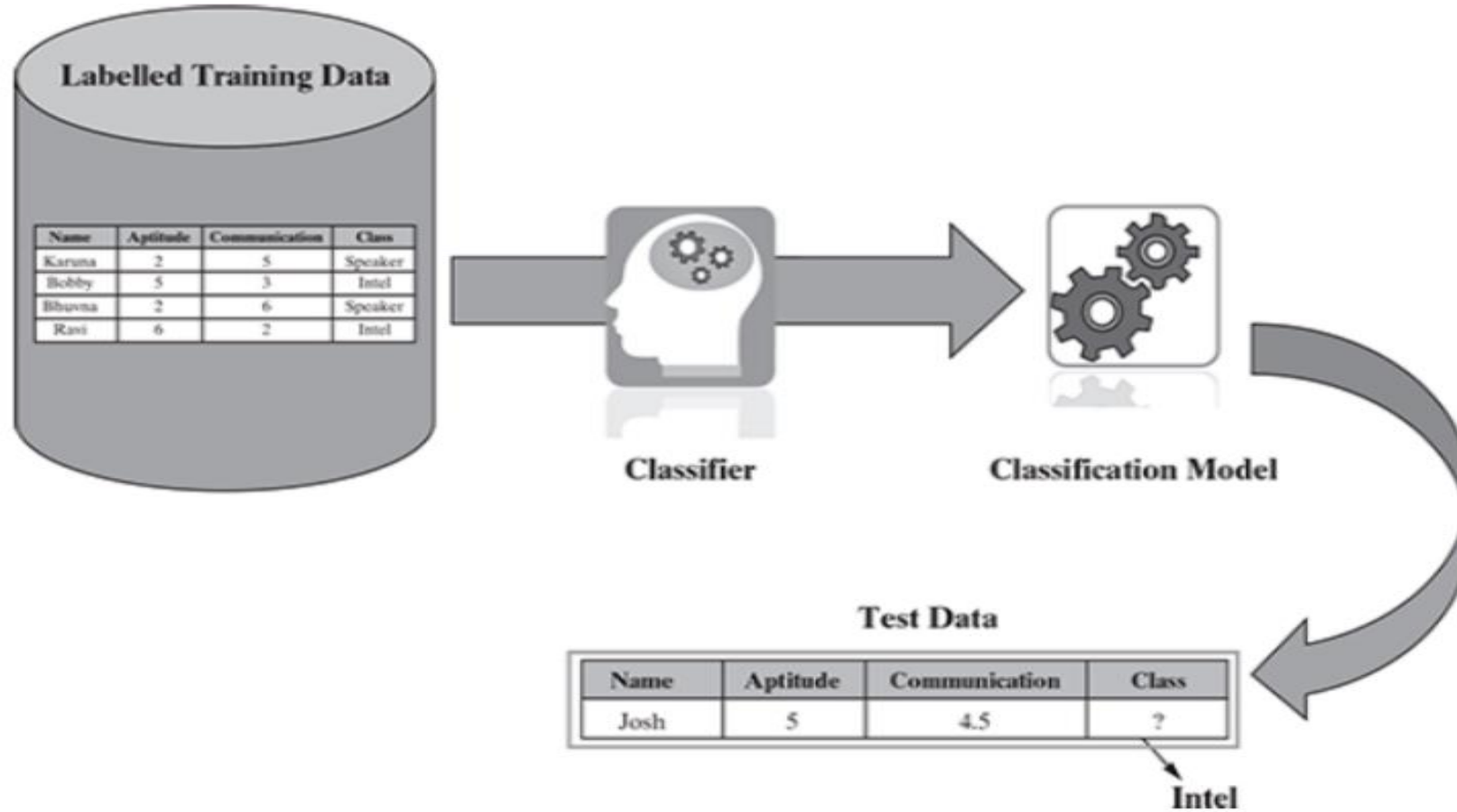
Examples of Supervised learning

- Predicting the results of a game.
- Predicting whether a tumor is malignant or benign.
- Predicting the price of domains like real estate, stocks, etc.
- Classifying texts such as classifying a set of emails as spam or no spam.
- When we are trying to predict a categorical or nominal variable, the problem is known as a **classification problem**. Whereas when we are trying to predict a real valued variable, the problem falls under the category of **regression**

Classification

- Classification is a type of supervised learning where a target feature, which is of type categorical, is predicted for test data based on the information imparted by training data. The target categorical feature is known as class.
- **Applications**
 - Image classification
 - Prediction of disease
 - Win-loss prediction of games
 - Prediction of natural calamity like earthquake, flood, etc.
 - Recognition of handwriting

Classification Process



- There are number of popular machine learning algorithms which help in solving classification problems.
- Naïve Bayes
- Decision tree
- k Nearest Neighbor algorithms

Regression

- In **linear regression**, the objective is to predict numerical features like real estate or stock price, temperature, marks in an examination, sales revenue, etc.
- The underlying predictor variable and the target variable are continuous in nature.
- Figure (next slide) shows a typical simple regression model, where regression line is fitted based on values of target variable with respect to different values of predictor variable.

$$Y = a + bX$$

- A typical linear regression model can be represented in the form – where 'X' is the predictor variable and 'Y' is the target variable.

Regression Model

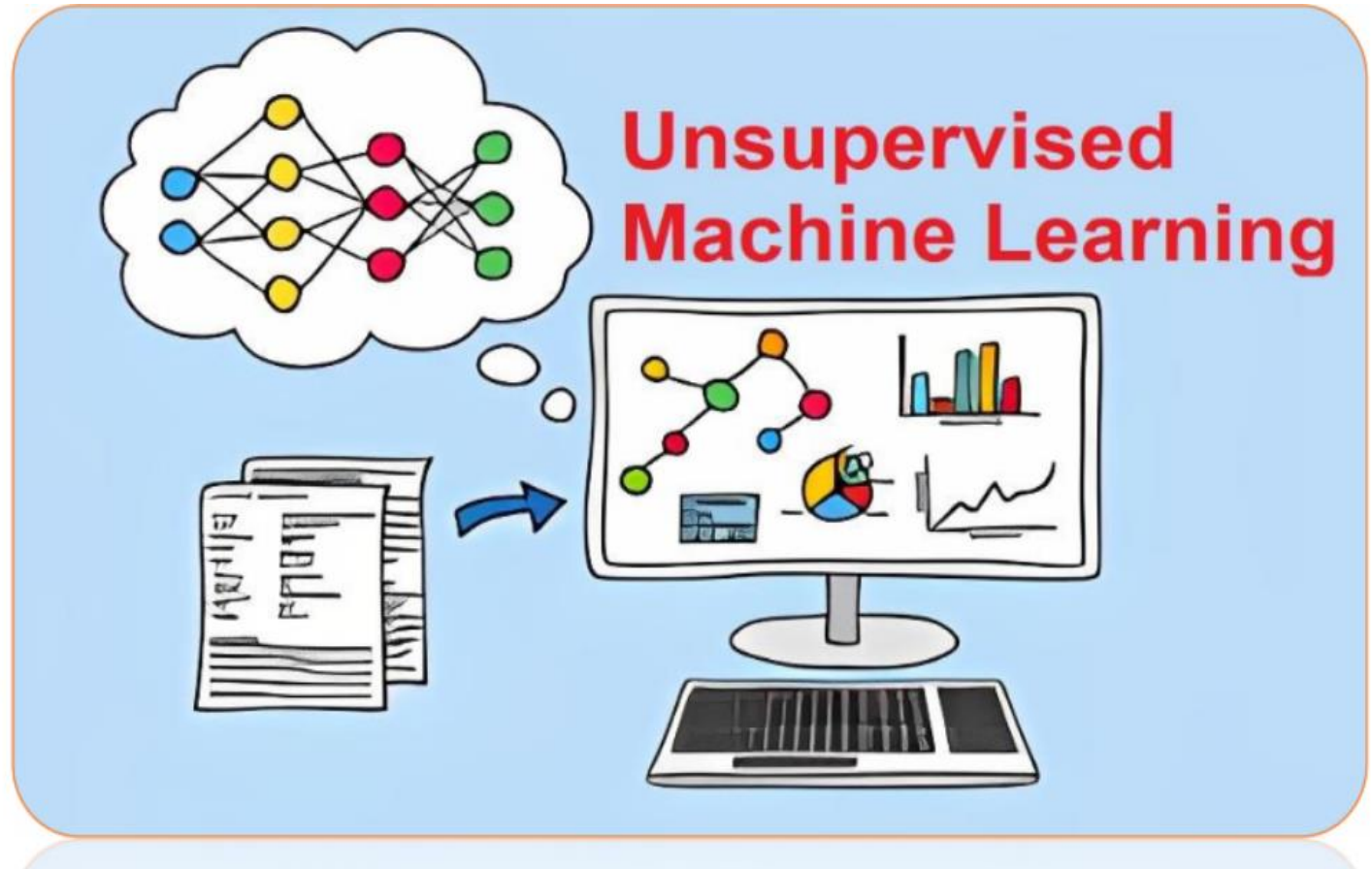


Typical applications of regression can be seen in

- Demand forecasting in retails
- Sales prediction for managers
- Price prediction in real estate
- Weather forecast
- Skill demand forecast in job market

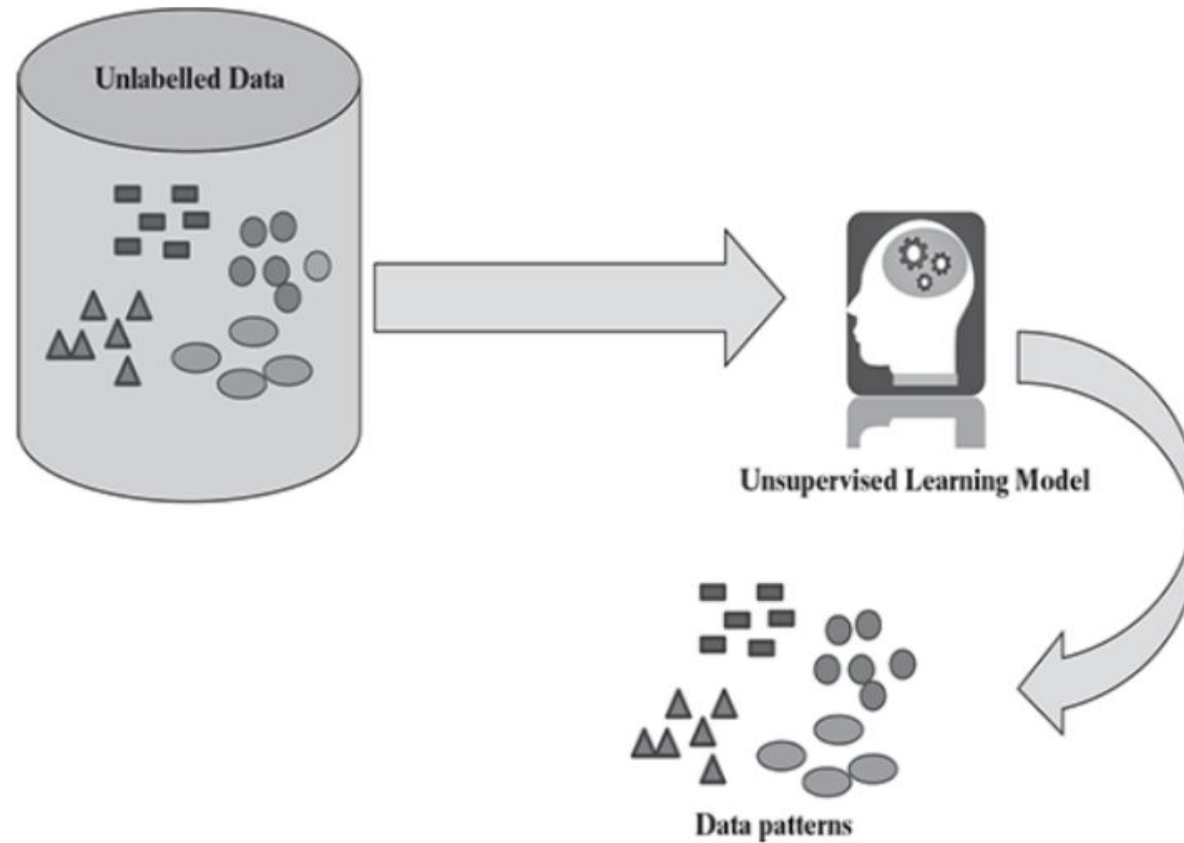
Unsupervised Learning

Unsupervised learning is a type of machine learning algorithm used to draw inferences from datasets consisting of input data without labelled responses.



- Unlike supervised learning, in unsupervised learning there is no labelled training data to learn from and no prediction to be made.
- In unsupervised learning, the objective is to take a dataset as input and try to find natural groupings or patterns within the data elements or records.
- Therefore, unsupervised learning is often termed as **descriptive model** and the process of unsupervised learning is referred as **pattern discovery or knowledge discovery**.
- One critical application of unsupervised learning is customer segmentation.

Unsupervised learning



Examples

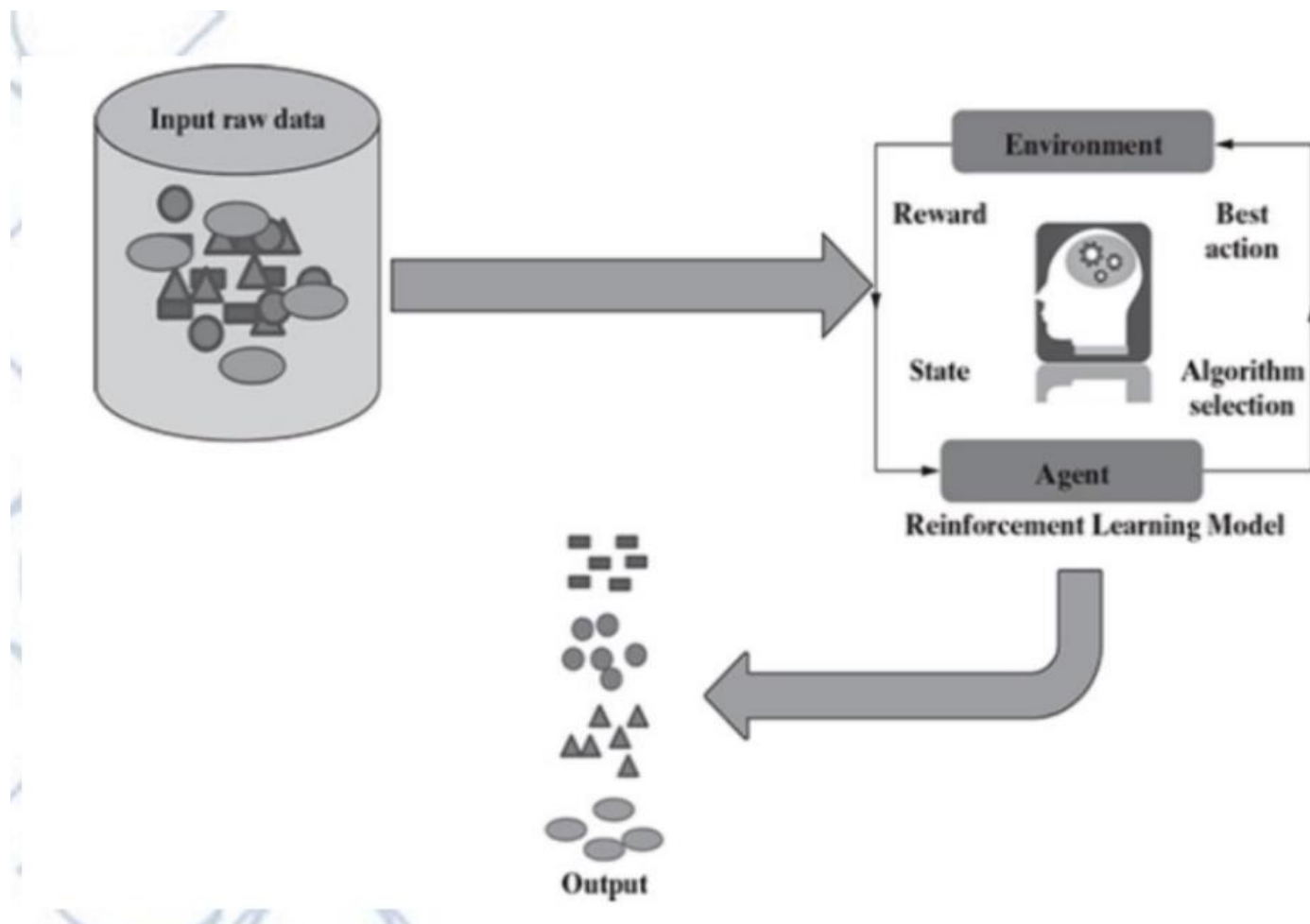
- Data exploration
- Customer segmentation,
- Recommender systems,
- Target marketing campaigns, and
- Data preparation and visualization, etc.
- **Clustering** is the main type of unsupervised learning. It intends to group or organize similar objects together.
- **Association analysis.** As a part of association analysis, the association between data elements is identified.

Reinforcement learning

Reinforcement Learning is a type of machine learning technique that enables an agent to learn in an interactive environment by trial and error using feedback from its own actions and experiences.



- Machines often learn to do tasks autonomously.
- Let's try to understand in context of the example of the child learning to walk. The action tried to be achieved is walking, the child is the agent and the place with hurdles on which the child is trying to walk resembles the environment.
- It tries to improve its performance of doing the task. When a sub-task is accomplished successfully, a reward is given.
- When a sub-task is not executed correctly, obviously no reward is given. This continues till the machine is able to complete execution of the whole task. This process of learning is known as reinforcement learning.



Applications

- One contemporary example of reinforcement learning is self-driving cars.
- The critical information which it needs to take care of are speed and speed limit in different road segments, traffic conditions, road conditions, weather conditions, etc.
- The tasks that have to be taken care of are start/stop, accelerate/decelerate, turn to left / right, etc.

Industry automation with Reinforcement Learning

- Reinforcement Learning applications in trading and finance
- Reinforcement Learning in NLP (Natural Language Processing)
- Reinforcement Learning applications in healthcare
- Reinforcement Learning in news recommendation
- Reinforcement Learning in gaming
- Real-time bidding—Reinforcement Learning applications in marketing and advertising
- Intelligent robots

Applications of M.L

- Machine learning has been adopted by various industry domains such as Banking and Financial Services, Insurance, Healthcare, Life Sciences, etc. to solve problems.
- Some of the most adopted platforms to implement machine learning include Python, R, MATLAB, SAS, SPPSS, etc.
- To avoid ethical issues, the critical consideration is required before applying machine learning and using any outcome from machine learning.

References

1. Machine Learning, Saikat Dutt, S. Chjandramouli, Das, Pearson
2. Machine Learning, Anuradha Srinivasaraghavan, Vincy Joseph, Wiley

Thank you!